

HOW CHEMICAL CHANGES TAKE PLACE

Physical Science
Quarter 3

Chemical Equation

- In chemistry, a chemical equation is needed to find the amount of materials (**reactants**) that you need and the new substance (**product**) to be formed.



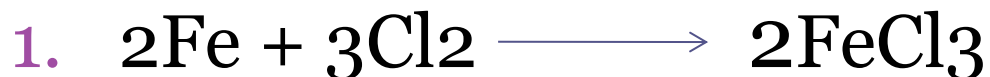
Examples

- Milk + Eggs + pancake mix $\longrightarrow$ Pancake
- 2 cups of rice + 2 cups of water $\longrightarrow$ 4 cups boiled rice

Law of Conservation of Matter

- **During a chemical reaction, atoms cannot be created nor destroyed.**
- Atoms of reactants are just rearranged to form the products.
- No new atoms enter the reaction and no original atoms disappear.
- In a chemical reaction, atoms are conserved.

Examples of Chemical Reaction



(iron)

(chlorine)

(iron (III) chloride)

Reactant side

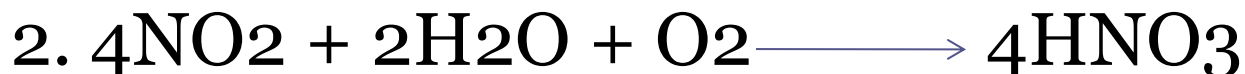
Product side

2Fe

2Fe

6Cl

6Cl



(nitrogen dioxide)

(water)

(oxygen)

(nitric acid)

Reactant side

Product side

4N

4N

12O

12O

4H

4H

Evaluation

- Count the number of atoms in the reactant side and the product side of the equation



Answers:



Reactant Side

4 C

8 H

12 O

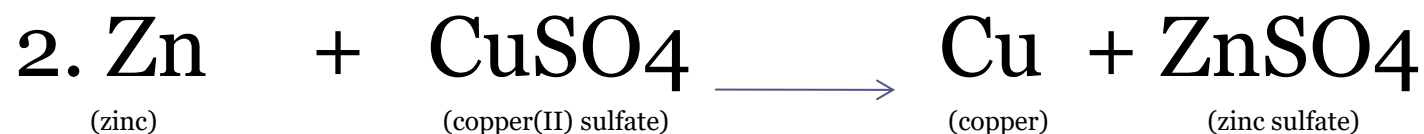
Product Side

4C

8H

12O

Answers:



Reactant Side

1Zn

1Cu

1S

4O

Product Side

1Zn

1Cu

1S

4O

Answers:



Reactant Side

1Ba

1S

4O

2Na

2Cl

Product Side

1Ba

1S

4O

2Na

2Cl

Balancing a Chemical Equation

- The number of atoms of an element in the reactant side and the product side must be equal.



Hints for Balancing Chemical Equations:

1. Count atoms of each element on the two sides of the equation.
2. Determine which atoms need to be balanced.
3. Balance element one at a time by placing the correct coefficient in front of the chemical formula.
4. Recheck the number of atoms in the two sides of the equation. Repeat if there are discrepancies in the result.

Answer (Exercise)



Reactant side

2H

2Cl

Product Side

2H

2Cl

One more Example



Reactants

2S

2O

Product

1S

2O



Reactants

2S

4O

Product

2S

4O

Evaluation

- Balance the following equations:

